

✓ MODULE–WISE QUESTION BANK (Java EE, JDBC, JSP, Servlets, Hibernate)

M1 – Introduction to Java EE

1. What is Java EE? Explain its architecture.
 2. Differentiate between Java SE and Java EE.
 3. What is the role of JDBC in web applications?
 4. What is the role of JSP in web applications?
 5. What is the role of Servlets in web applications?
 6. Compare the roles of JDBC, JSP, and Servlet in a web application.
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M2 – JDBC (Java Database Connectivity)

1. What is JDBC? Explain its need.
 2. Explain different types of JDBC drivers.
 3. Explain JDBC architecture with a neat diagram.
 4. Explain the steps to connect a Java program to a database.
 5. How do you execute SELECT, INSERT, UPDATE, and DELETE queries in JDBC?
 6. Differentiate between Statement and PreparedStatement.
 7. What is ResultSet? Explain its types.
 8. What is Metadata? Explain DatabaseMetaData and ResultSetMetaData.
 9. Explain transaction management in JDBC.
 10. What is commit, rollback, and savepoint?
 11. Explain batch processing in JDBC with an example.
 12. Discuss SQL injection and how PreparedStatement prevents it.
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M3 – Java Server Pages (JSP)

1. What is JSP? Explain its advantages.
2. Explain the JSP lifecycle.
3. What are JSP scriptlet, expression, and declaration?
4. What are JSP implicit objects? Explain any five.
5. Explain JSP directives with examples.
6. What are JSP action elements? Explain any three.

7. Explain how JavaBeans are used in JSP.
 8. What is JSP Expression Language (EL)? Explain its features.
 9. Explain JSTL core tags: <c:out>, <c:set>, <c:if>, <c:choose>, <c:when>, <c:otherwise>, <c:forEach>.
 10. Explain JSTL function tags.
 11. What is a custom tag library? Explain with example.
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M4 – Servlets

1. What is a Servlet? Explain its advantages.
 2. Explain the servlet lifecycle with diagram.
 3. Explain GET and POST request handling in servlets.
 4. What is Servlet Deployment Descriptor (web.xml)? Explain its purpose.
 5. What is ServletContext? What are its uses?
 6. What is ServletConfig? How is it different from ServletContext?
 7. Explain session management techniques in servlets.
 8. What are cookies? Explain how to create and read cookies in a servlet.
 9. What is servlet chaining? Explain with example.
 10. What are servlet filters? Explain their purpose.
 11. Write the steps to upload and download files using servlets.
 12. Explain request dispatching in servlets.
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M5 – JSP and Servlet Integration

1. Differentiate between JSP and Servlet.
 2. Explain how JSP and Servlets work together in a web application.
 3. Explain the Model–View–Controller (MVC) architecture.
 4. Explain request forwarding between JSP and Servlet.
 5. Explain the role of MVC in web application design.
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M6 – Hibernate Framework

1. What is Hibernate? Explain its features.
2. What are the advantages of ORM over JDBC?

3. Explain Hibernate architecture and core components.
4. Explain the steps to set up Hibernate in a Java application.
5. What is mapping? How do you map Java classes to database tables?
6. Explain Hibernate configuration using XML.
7. Explain Hibernate configuration using annotations.
8. Explain basic CRUD operations in Hibernate.
9. Explain the basics of Hibernate Query Language (HQL).
10. Differentiate between SQL and HQL.